ERP Reports — Complete Workflow & Wireframe Specification

ERP-DEV-07 • Route: /erp/reports • Primary Page: ERPReportsPage.tsx

Purpose. This document defines the complete functional behavior and UI structure required to build the ERP Reports page. It converts the supplied TypeScript domain models and ERP-DEV-07 ownership requirements into an implementation-ready workflow and wireframe specification.

Scope. The page is responsible for supply-chain reporting and analysis: report filtering, inventory valuation, vendor delivery performance audit, carrier turnaround analytics, export preview, and presentation of report data that can also feed ERP Dashboard summaries.

> **Important boundary.** The Reports page reads ERP transaction data and analytics. It must not modify core transaction forms such as Purchase Orders, Sales Orders, or Inventory Adjustments.

1. Module Context

ERP Reports is an analytical layer over operational ERP data. The page should make it possible for an authorized user to select a reporting period and business scope, run one or more reports, inspect the results, understand exceptions, and export the resulting dataset without navigating through transaction-entry screens.

- **Route:** /erp/reports
- **Primary component:** ERPReportsPage.tsx
- **Developer ownership:** ERP-DEV-07
- **Primary responsibility:** supply-chain report filters, inventory valuation, vendor delivery audit, carrier turnaround analytics, export preview tables.
- **Read-only behavior:** reports should not create, update, approve, cancel, or delete ERP transactions.
- **Cross-team dependency:** ERP-DEV-01 Dashboard consumes/aggregates report-derived summary information.

2. Canonical Data Model and Reporting Meaning

The five supplied interfaces establish the minimum reporting data model. The Reports page should treat these objects as source records and derive analytical values from them rather than duplicating transaction-editing logic.

InventoryItem represents stock position at a product/warehouse level. The report layer can use quantity on hand, reserved quantity, available quantity, in-transit quantity, average unit cost, last stock-take date, and reorder point to explain inventory value, availability, and replenishment risk.

PurchaseOrder provides procurement order dates, expected delivery dates, status, currency, financial totals, vendor relationship, and PO items. It supports vendor delivery analysis, purchasing spend summaries, and timing calculations.

SalesOrder provides customer order dates, delivery due dates, status, financial totals, payment status, and order items. It supports outbound demand and fulfillment-related reporting and can be used when report filters need to connect supply movement with sales demand.

StockMovement records inventory movement type, quantity, source and destination warehouses, reference document, movement date, and performer. It is the primary source for movement history and turnaround

calculations.

VendorEvaluation stores quality, delivery, pricing, and overall scores. It supports vendor performance context and can be displayed alongside delivery audit results.

Important calculation principle: report calculations should be derived consistently from these source fields. For example, available stock should not be independently invented by the UI when *quantityAvailable* already exists.

3. ERP Reports Page — Overall Workflow

The page follows a controlled report lifecycle so that users always know whether they are configuring a report, generating it, reviewing results, or exporting a snapshot.

Step 1 — Open page. The application loads /erp/reports and establishes the default reporting context. The user sees the page header, report navigation, filter panel, report cards/tabs, and an initial empty-state or recent/default report view.

Step 2 — Define scope. The user selects report type, date range, warehouse/vendor/carrier/product scope where applicable, status, and other supported filters. Required fields are validated before execution.

Step 3 — Run report. Selecting Generate/Run Report sends the current filter state to the reporting layer. A loading state is shown and the report result is generated without changing transactional records.

Step 4 — Inspect summary. The report displays high-level KPIs first so the user can immediately understand the selected period. Examples include total inventory value, delivery performance, late PO count, average turnaround time, and vendor score.

Step 5 — Inspect details. The user moves into detailed sections containing product-level, warehouse-level, vendor-level, movement-level, or carrier-level records. Sorting, search, pagination, and drill-down behavior should preserve the active filter context.

Step 6 — Preview export. The user chooses Export. Instead of immediately downloading, the page opens an export preview showing the selected report, active filters, visible columns, record count, and output format.

Step 7 — Export. The user confirms the format and downloads the generated snapshot. Export must represent the same filter state and result set currently displayed.

Step 8 — Dashboard aggregation. Summary values that are required by ERP-DEV-01 can be exposed through the reporting/data layer so the Dashboard can present consistent ERP metrics.

4. Primary Page Layout / Wireframe

The following wireframe is a conceptual implementation blueprint rather than a screenshot. It describes the visual hierarchy and the information that should exist on the actual React page.

```
■■■
```

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ FILTERS                                                                      │
│ Date Range [__________] [__________] Warehouse [All ▼] Vendor [All ▼]        │
│ Product [All ▼] Status [All ▼] Search [____________________]                 │
│ [Reset] [Run Report]                                                         │
└──────────────────────────────────────────────────────────────────────────────┘
┌──────────────────────────────────────────────────────────────────────────────┐
│ KPI SUMMARY                                                                  │
│ [KPI] [KPI] [KPI] [KPI]                                                      │
└──────────────────────────────────────────────────────────────────────────────┘
┌──────────────────────────────────────────────────────────────────────────────┐
│ REPORT VISUAL / ANALYTICAL AREA                                              │
│                                                                              │
│ chart / trend / distribution exception / insights panel                      │
│                                                                              │
└──────────────────────────────────────────────────────────────────────────────┘
┌──────────────────────────────────────────────────────────────────────────────┐
│ DETAIL RESULTS                                                               │
│ Search | Columns | Sort | Page size                                          │
│ ┌──────────────────────────────────────────────────────────────────────────┐ │
│ │ Result rows / expandable details / pagination                            │ │
└──────────────────────────────────────────────────────────────────────────────┘
```

The top-level layout should remain stable while the report-specific content changes below the filter area. This allows users to switch between reports without relearning the page.

5. Report 1 — Inventory Valuation

The Inventory Valuation report explains the monetary value and physical position of inventory across products and warehouses. Its core calculation is based on inventory quantity and average unit cost.

> **Recommended calculation:** Inventory Value = quantityOnHand × averageUnitCost. A second analytical view can separately show available value using quantityAvailable × averageUnitCost, but it should be explicitly labeled so users do not confuse physical stock value with immediately available stock value.

- **Primary filters:** date/report snapshot context, warehouse, product, product category if available, stock status, low-stock/reorder condition.
- **Summary:** total quantity on hand, total available quantity, total reserved quantity, total in-transit quantity, total inventory value, number of items below reorder point.
- **Detail:** product, warehouse, quantity on hand, reserved, available, in transit, average unit cost, calculated value, reorder point, last stock-take date.
- **Exception view:** items where available quantity is low or where quantity on hand is at/below reorder point.
- **Drill-down:** selecting an inventory row can open its movement history or related stock activity in a read-only context.

6. Report 2 — Vendor Delivery Performance Audit

This report evaluates whether purchase orders associated with vendors were delivered according to the expected delivery date. It should clearly distinguish scheduled delivery from actual movement/receipt evidence available to the reporting layer.

> **Core timing concept:** delivery variance can be represented as Actual Delivery Date – Expected Delivery Date. Positive variance indicates delivery after the expected date; negative variance indicates delivery before the expected date. If actual delivery evidence is unavailable, the report must show an explicit 'Actual delivery date unavailable' state rather than silently calculating a result.

- **Filters:** vendor, PO status, order date range, expected delivery date range, currency, performance band, product/warehouse scope where supported.
- **Summary:** total POs evaluated, delivered POs, late POs, on-time rate, average delivery variance, vendor evaluation score where available.
- **Detail:** PO number, vendor, order date, expected delivery date, actual delivery evidence, delivery variance, PO status, total amount, evaluation scores.
- **Audit behavior:** allow the user to trace the report row back to the PO reference without providing transaction editing controls.

7. Report 3 — Carrier Turnaround Time Analytics

> Carrier turnaround analysis measures elapsed time associated with shipment or stock movement processes. Because the supplied StockMovement interface does not contain an explicit carrierId or carrier timestamps, the UI must not pretend that those fields exist.

If the application already has carrier/shipment data elsewhere, ERP Reports can join that data through the reporting service. If not, the first implementation should expose only the turnaround metrics that can be reliably derived from available movement/reference timestamps.

- **Potential metrics when carrier timestamps exist:** average turnaround time, median turnaround time, fastest/slowest turnaround, delayed shipments, turnaround by carrier, warehouse-to-warehouse turnaround.
- **Available StockMovement context:** movementCode, movementType, quantity, fromWarehouseId, toWarehouseId, referenceType, referenceId, movementDate, performedBy.
- **Fallback behavior:** if carrier-specific records are not available, present movement turnaround analytics as 'Stock Movement Turnaround' rather than fabricating carrier analytics.

8. Report 4 — Stock Movement Analysis

Although the primary ownership statement emphasizes inventory valuation, vendor delivery, carrier turnaround, and export preview, StockMovement is a canonical type and therefore should have a supporting analytical view or drill-down.

- Movement volume by movement type.
- Quantity moved by warehouse and direction.
- Purchase-order, sales-order, transfer, and adjustment references.
- Movement trend by date.
- Inbound versus outbound movement where the movement semantics allow that classification.
- Read-only movement detail with performer and remarks.

9. Vendor Evaluation Context

VendorEvaluation should enrich vendor reports rather than replace operational vendor records. The page can display qualityRating, deliveryRating, pricingRating, overallScore, evaluator, evaluationDate, and remarks when the selected vendor has evaluation data.

A useful UX pattern is to place vendor evaluation context beside delivery audit results: delivery performance answers what happened operationally, while evaluation scores provide the organization's recorded assessment. The page should keep these concepts visually distinct.

10. Filter Panel — Detailed Behavior

The filter panel is the control center of ERP Reports. It should be persistent and visible above report results. Filters should be context-sensitive: selecting Inventory Valuation should show inventory-related controls, while Vendor Delivery should expose vendor/PO controls.

- **Date range:** supports start and end dates; start must not be later than end.
- **Warehouse:** single or multi-select depending on backend support; default can be All Warehouses.
- **Vendor:** enabled for vendor/PO reports; disabled or hidden when irrelevant.
- **Product:** enabled for inventory and movement reports.
- **Status:** report-specific statuses such as PO/SO/inventory status.
- **Search:** free-text search across the current report's supported identifiers.
- **Reset:** returns filters to defined defaults without changing the user's current report type.
- **Run Report:** executes with the exact visible filter state.
- **Filter persistence:** switching between report types should preserve compatible filters and clear incompatible filters.

11. Results Area and User Interaction

The results area should follow a consistent sequence: KPI summary → analytical visualization → exceptions/insights → detailed records. This prevents users from having to scan a large table before understanding the report.

KPI cards. Each card should have a concise label, current value, optional comparison context, and a tooltip or secondary description where the calculation needs clarification.

Visualization. Use charts only when they communicate trends or distributions better than text. Inventory reports can show value by warehouse or stock status; vendor reports can show on-time versus late performance; movement reports can show quantity over time.

Exception panel. Surface actionable anomalies such as low-stock items, late purchase orders, missing delivery evidence, or unusually high turnaround time. The panel should link to the underlying result rows.

Detail table. The table is the evidence layer. It should support search, sorting, pagination, column visibility, and row expansion where appropriate. Numeric values should be formatted consistently with the selected currency and unit conventions.

12. Loading, Empty, Error and Permission States

Initial empty state: Explain what the user can report and provide a clear Run Report action.

Loading: Keep filter controls understandable, show skeleton KPI cards and result placeholders, and prevent duplicate report execution while the request is running.

No results: State that no records match the selected filters and provide Reset Filters / Adjust Filters actions.

Partial data: If a report depends on missing fields such as actual delivery timestamp or carrier identity, explicitly identify which metric could not be calculated.

Error: Show a user-readable failure message, preserve the selected filters, and provide Retry.

Permission: If the user is not authorized to view financial or supply-chain data, the page should follow platform RBAC and avoid exposing restricted values through exports, tooltips, or hidden DOM content.

13. Export Preview Workflow

Export is treated as a controlled reporting action rather than a direct download. This ensures the user knows exactly what is being exported.

- User selects Export.
- System opens Export Preview.
- Preview displays report name, date range, active filters, record count, selected columns, and a compact sample of rows.
- User selects output format supported by the application, such as CSV or XLSX.
- User confirms export.
- System generates the file from the same report query/result context.
- System shows success feedback and preserves the report screen.

Export integrity rule: the export must not silently ignore active filters. If the export is limited to visible columns or current page rows, that scope must be explicitly stated.

14. Dashboard Dependency — ERP-DEV-01 ↔ ERP-DEV-07

ERP-DEV-01 owns the ERP Dashboard while ERP-DEV-07 owns Reports. The dependency should be handled through a shared reporting/data contract rather than copying calculations between pages.

- Reports provides authoritative report metrics and aggregation definitions.
- Dashboard consumes approved summary metrics for cards, charts, and trend widgets.
- Both pages should use the same date/filter semantics and calculation definitions.
- A dashboard KPI should link to the corresponding report with the relevant filter context when drill-down is supported.
- Changes to report calculation logic should be communicated because they can change dashboard summaries.

15. Frontend Component Structure

A maintainable React implementation can separate page orchestration, filter state, report-specific views, and export behavior.

```
src/features/erp/reports/
■■■ ERPReportsPage.tsx
■■■ components/
■ ■■■ ReportHeader.tsx
■ ■■■ ReportTypeSelector.tsx
■ ■■■ ReportFilterPanel.tsx
■ ■■■ ReportKpiCards.tsx
■ ■■■ ReportChart.tsx
■ ■■■ ReportExceptions.tsx
■ ■■■ ReportResultsTable.tsx
■ ■■■ ExportPreview.tsx
■ ■■■ ReportEmptyState.tsx
■■■ inventory/
■ ■■■ InventoryValuationReport.tsx
■■■ vendors/
■ ■■■ VendorDeliveryPerformanceReport.tsx
■■■ logistics/
■ ■■■ CarrierTurnaroundReport.tsx
■■■ movements/
■ ■■■ StockMovementReport.tsx
■■■ hooks/
■ ■■■ useERPReports.ts
■■■ types/
■■■ reportTypes.ts
```

The exact folder names can follow the project's existing conventions. The important architectural rule is that ERPReportsPage orchestrates state and report selection while report-specific components remain isolated.

16. State Model for ERPReportsPage

The page should maintain explicit UI state so report execution, loading, errors, and export preview are deterministic.

```
reportType: 'INVENTORY_VALUATION' | 'VENDOR_DELIVERY' |
'CARRIER_TURNAROUND' | 'STOCK_MOVEMENT'
filters: { dateFrom, dateTo, warehouseIds, vendorIds,
productIds, status, search }
status: 'IDLE' | 'LOADING' | 'SUCCESS' | 'ERROR'
results: report-specific result data
selectedColumns: string[]
exportPreviewOpen: boolean
exportFormat: 'CSV' | 'XLSX' | null
errorMessage: string | null
```

Separating **filters** from **results** is important. A user can change a filter while the previous result remains visible until the next report execution, or the application can intentionally clear the result when a filter becomes dirty. Either behavior should be chosen consistently.

17. Validation and Business Rules

- Date From cannot be later than Date To.
- A report cannot execute while mandatory filter values are invalid.
- Numeric quantities must not be displayed with inconsistent signs or units.
- Currency values must use the report's currency context; multi-currency aggregation requires explicit conversion rules.
- Missing actual delivery timestamps must not be interpreted as on-time delivery.
- Missing vendor evaluations must be represented as unavailable, not zero.
- A zero score and a missing score are different states.
- Report filters must not mutate source transaction objects.
- Exports must respect the same authorization rules as on-screen data.
- Long-running report requests should support cancellation or at minimum prevent duplicate submissions.

18. Detailed End-to-End User Journey

Scenario A — Inventory manager reviews stock value. The user opens ERP Reports, chooses Inventory Valuation, selects a warehouse and reporting context, runs the report, reviews total value and low-stock exceptions, expands a product row, checks the quantity/cost components, and exports the filtered result.

Scenario B — Procurement manager audits vendor delivery. The user selects Vendor Delivery Performance, chooses a vendor and period, runs the report, reviews late POs and delivery variance, opens a PO reference for read-only traceability, compares the operational delivery result with VendorEvaluation scores, and exports the audit.

Scenario C — Operations user analyzes turnaround. The user selects Carrier Turnaround when carrier/shipment data is available. The page displays turnaround metrics and delayed cases. If carrier identity is unavailable in the underlying data contract, the page clearly switches to supported stock-movement turnaround analysis rather than inventing carrier values.

19. Recommended Visual Hierarchy

The page should feel like an enterprise analytics workspace rather than a transaction form. Use a clean report header, compact filter controls, strong KPI hierarchy, restrained charts, and a dense but readable evidence table.

- Header: page title + description + Refresh + Export.
- Report selector: clear active report state.
- Filters: grouped and visually separated from results.
- KPIs: 4–6 high-value metrics, not a large collection of decorative cards.
- Analytics: one primary chart plus supporting exception/insight content.
- Table: optimized for operational reading with sticky headers and pagination.
- Export preview: modal/drawer with exact scope confirmation.

20. Final Implementation Acceptance Criteria

- Route /erp/reports loads ERPReportsPage.tsx.
- User can select a report type and run report-specific filters.
- Inventory Valuation uses InventoryItem fields and exposes clear valuation calculations.
- Vendor Delivery Performance uses PurchaseOrder data and clearly handles unavailable actual-delivery evidence.
- Carrier Turnaround does not fabricate carrier-specific fields absent from the supplied data model.
- StockMovement can be inspected analytically or through report drill-down.
- VendorEvaluation scores are presented as evaluation context and not confused with raw delivery metrics.
- Loading, empty, error, and permission states are implemented.
- Export Preview shows active filters, result count, selected columns, and output format.
- Exports preserve report scope and authorization.
- Reports remain read-only and do not modify PO, SO, or Inventory Adjustment forms.
- Dashboard metrics can consume shared report definitions through the ERP-DEV-01 / ERP-DEV-07 dependency.

21. Final Wireframe Summary

The finished ERP Reports page should communicate one simple flow: **Choose report** → **Define filters** → **Run** → **Understand KPIs** → **Inspect analytics** → **Investigate exceptions** → **Review detailed evidence** → **Preview export** → **Export.**

The page is therefore not just a collection of tables. It is an analytical workflow that connects operational ERP records to decision-ready reporting while preserving traceability, calculation transparency, permissions, and read-only separation from transaction workflows.

Document basis: supplied InventoryItem, PurchaseOrder, SalesOrder, StockMovement, VendorEvaluation interfaces and ERP-DEV-07 ownership/dependency requirements.